Ex.No. 3.2 – Design and Draft the Given 3D Component

AIM

To create the 3D model and dimensioned drawing of the given arched support bracket using Autodesk Fusion 360.

PROCEDURE

1. Open Autodesk Fusion 360, create a new design, and set the document units to millimetres.

2. On the XY plane, sketch the base plate with an overall length of 420 mm and width of 150 mm. Form the rounded ends and extrude the plate to a thickness of 30 mm.

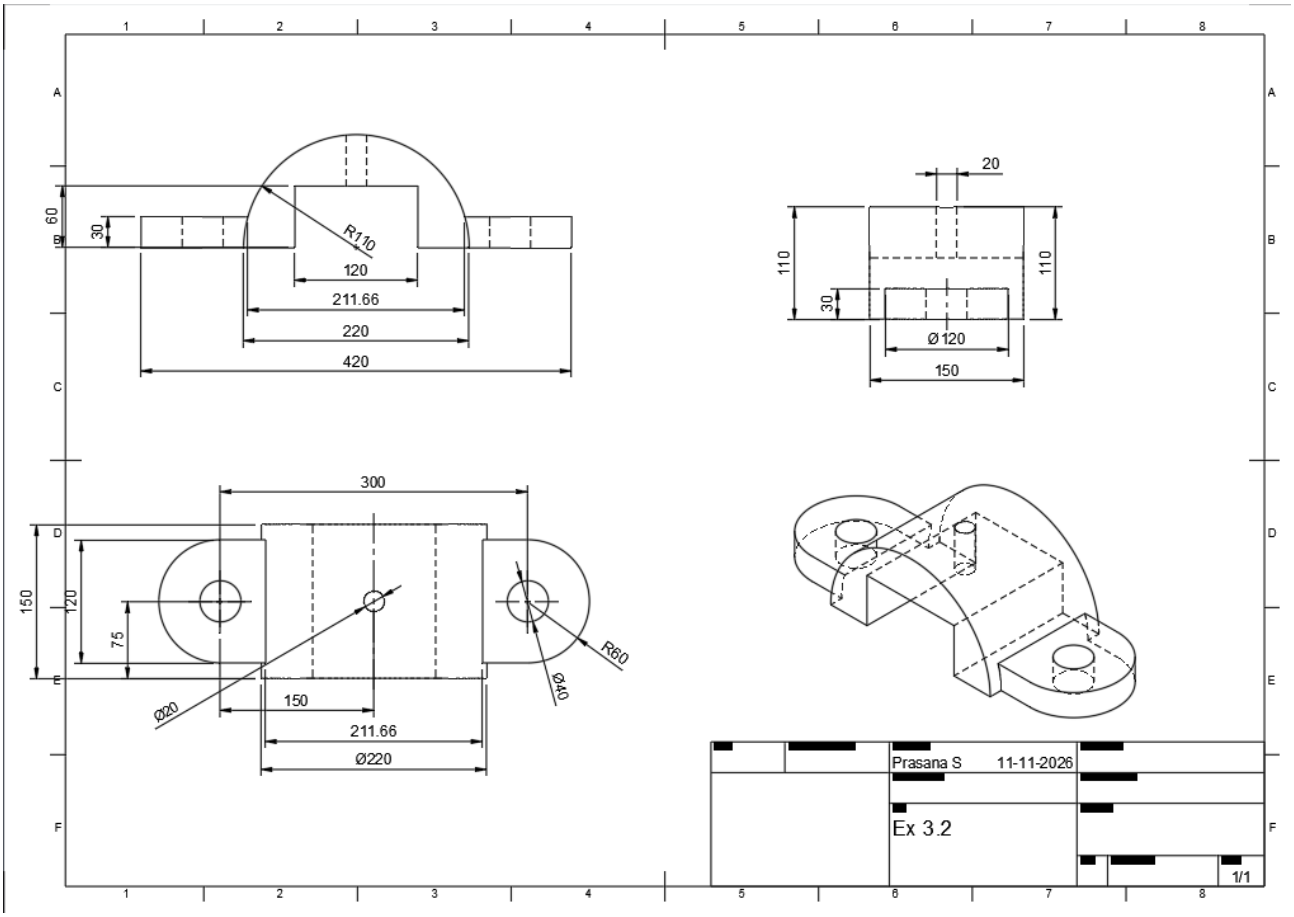
3. Create two mounting holes of Ø40 mm on the rounded ends of the base plate. Position the hole centres symmetrically, 75 mm from each end, and cut them through the plate.

4. On the central region of the base, sketch the arched support profile. Use an outer radius of R110 mm and form the inner opening with a width of 120 mm. Keep the arch span at 220 mm and extrude the profile across the 150 mm width.

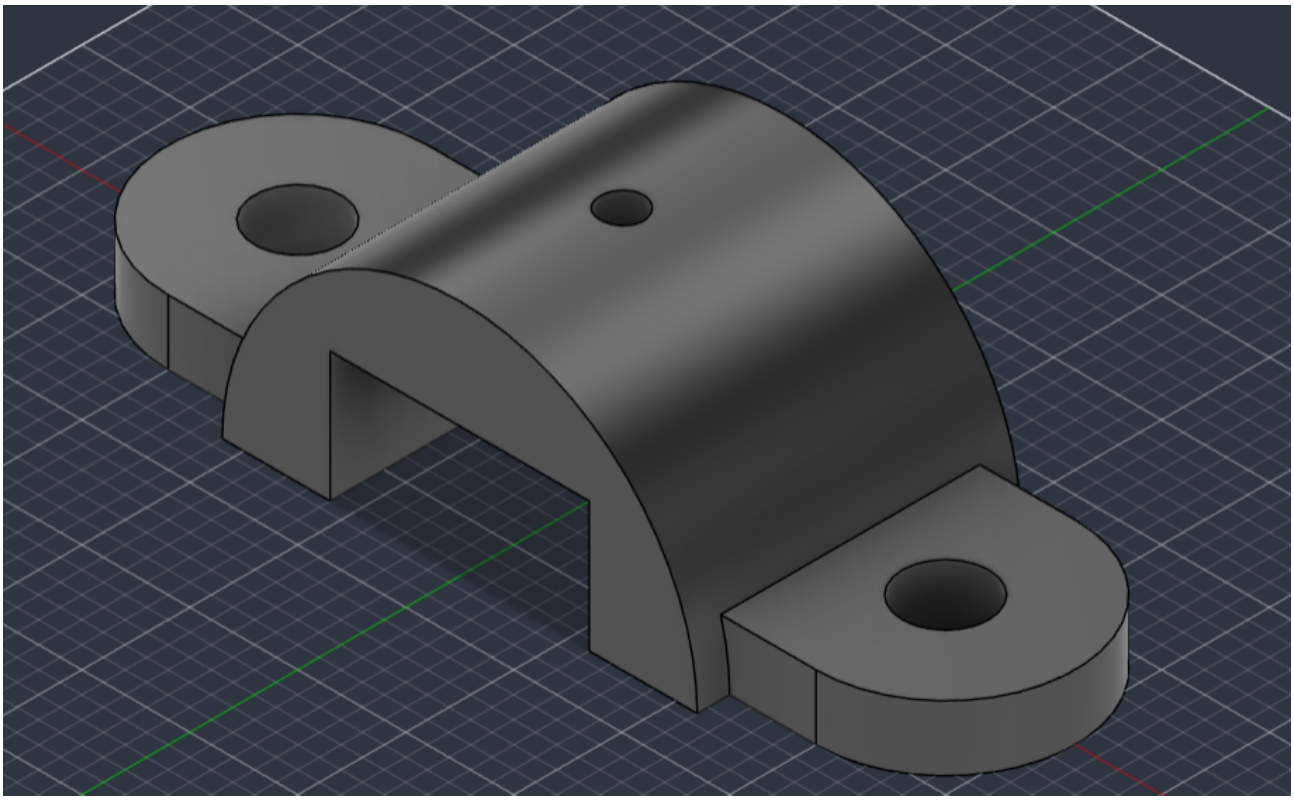
5. Create the central hole of Ø20 mm on the upper curved surface at the location shown in the drawing. Use the Hole or Extrude Cut command to make the opening.

6. Apply the required geometric constraints and verify the main dimensions: 420 × 150 mm base, 30 mm base thickness, Ø40 mm mounting holes, R110 mm outer arch, 220 mm arch span, 120 mm inner opening, and Ø20 mm top hole. Save the model and prepare the dimensioned drawing.

SKETCHES



OUTPUT



RESULT

Thus, the given arched support bracket was modelled and drafted using Autodesk Fusion 360.